

PROJECT REPORT

RICE PRODUCTION PREDICTION USING LONG SHORT-TERM MEMORY (LSTM), GATED RECURRENT UNIT (GRU), AND BIDIRECTIONALLSTM (BILSTM)

Putra Ramanda Yudhistira

22.K1.0028

Faculty of Computer Science
Soegijapranata Catholic University
2026

APPROVAL AND RATIFICATION PAGE

Judul Tugas Akhir	RICE PRODUCTION PREDICTION USING LONG SHORT-TERM MEMORY (LSTM), GATED RECURRENT UNIT (GRU), AND BIDIRECTIONALLSTM (BILSTM)
Diajukan oleh	PUTRA RAMANDA YUDHISTIRA
NIM	22.K1.0028
Tanggal disetujui	23 April 2026
Telah setujui oleh	
Pembimbing	Dr. SHINTA ESTRI WAHYUNINGRUM S.Si., M.Cs
Penguji 1	YONATHAN PURBO SANTOSA S.Kom., M.Sc.
Penguji 2	ROSITA HERAWATI S.T., M.I.T.
Penguji 3	Dr. YULIANTO TEJO PUTRANTO S.T., M.T.
Penguji 4	Dr. SHINTA ESTRI WAHYUNINGRUM S.Si., M.Cs
Ketua Program Studi	YONATHAN PURBO SANTOSA S.Kom., M.Sc.
Dekan	Prof. Dr. F. RIDWAN SANJAYA S.E., S.Kom., MS.IEC.

Halaman ini merupakan halaman yang sah dan dapat diverifikasi melalui alamat di bawah ini.

sintak.unika.ac.id/skripsi/verifikasi/?id=22.K1.0028

DECLARATION OF AUTHORSHIP (Heading Plain)

I, the undersigned:

Name : PUTRA RAMANDA YUDHISTIRA

ID : 22.K1.0028

declare that this work, titled "RICE PRODUCTION PREDICTION USING LONG SHORT-TERM MEMORY (LSTM), GATED RECURRENT UNIT (GRU), AND BIDIRECTIONALLSTM (BILSTM)", and the work presented in it is my own. I confirm that:

1. This work was done wholly or mainly while in candidature for a research degree at Soegijapranata Catholic University
2. Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
3. Where I have consulted the published work of others, this is always clearly attributed.
4. Where I have quoted from the work of others, the source is always given.
5. Except for such quotations, this work is entirely my own work.
6. I have acknowledged all main sources of help.
7. Where the work is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Semarang, April 28,2026



PUTRA RAMANDA YUDHISTIRA

22.K1.0028

HALAMAN PERNYATAAN PUBLIKASI KARYA ILMIAH UNTUK KEPENTINGAN AKADEMIS

Yang bertanda tangan dibawah ini:

Nama : Putra Ramanda yudhistira

Program Studi : Teknik Informatika

Fakultas : Ilmu Komputer

Jenis Karya : Skripsi

Menyetujui untuk memberikan kepada Universitas Katolik Soegijapranata Semarang Hak Bebas Royalti Nonekslusif atas karya ilmiah yang berjudul "RICE PRODUCTION PREDICTION USING LONG SHORT-TERM MEMORY (LSTM), GATED RECURRENT UNIT (GRU), AND BIDIRECTIONALLSTM (BILSTM)". Dengan Hak Bebas Royalti Nonekslusif ini Universitas Katolik Soegijapranata berhak menyimpan, mengalihkan media/formatkan, mengelola dalam bentuk pangkalan data (database), merawat, dan mempublikasikan tugas akhir ini selama tetap mencantumkan nama saya sebagai penulis / pencipta dan sebagai pemilik Hak Cipta.

Demikian pernyataan ini saya buat dengan sebenarnya.

Semarang, 28 APRIL 2026

Yang menyatakan



PUTRA RAMANDA YUDHISTIRA

22.K1.0028

ACKNOWLEDGMENT

Silahkan tuliskan anda ingin mengucapkan terima kasih atau ucapan persembahan ke siapapun yang anda rasa perlu ditulis disini

I have received a myriad of support, advice, and assistance throughout this document writing. I would like to thank my supervisors for formulating this topic. I would also like to thank my friend ADONIS MAX BALINDA for guiding with advice to finish this document.

I would like to thank my family and friends for giving me ceaseless love, support, and advices throughout my study in Soegijapranata Catholic University. You gave me great escape to rest my mind from my thesis.

ABSTRACT (ABSTRACT TITLE)

Rice yield prediction is a crucial contributor to national food security in Indonesia. This study aims to formulate a national rice yield forecasting model using Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Bidirectional LSTM (BiLSTM) models. These three models were selected because of their ability to learn time series patterns and sequential data with long dependencies, with BiLSTM being able to process both time directions simultaneously. Multivariate annual time series data from 2000 to 2024 were used in this study. This dataset consists of rice production data from the Central Statistics Agency (BPS) and climate variables such as rainfall, and average temperature obtained from the BMKG and NOAA. The collected data underwent pre-processing, including normalization and quality control, and was then divided into training and testing data. The training process was conducted separately for each model. Model performance was evaluated using the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) metrics to measure prediction accuracy. The results of this study are expected to indicate which model is most optimal in predicting national rice production, while also evaluating the contribution of climate variables in improving predictions.accuracy.

Keywords: Rice production prediction, LSTM, GRU, BiLSTM, Climate

TABLE OF CONTENTS

COVER ..... **i**

APPROVAL AND RATIFICATION PAGE ..... **ii**

DECLARATION OF AUTHORSHIP (Heading Plain) ..... **iii**

HALAMAN PERNYATAAN PUBLIKASI KARYA ILMIAH UNTUK KEPENTINGAN AKADEMIS ..... **iv**

ACKNOWLEDGMENT ..... **v**

ABSTRACT (Abstract Title) ..... **vi**

TABLE OF CONTENTS ..... **vii**

LIST OF FIGURE ..... **ix**

LIST OF TABLE ..... **x**

CHAPTER 1 INTRODUCTION **1**

1.1. Background 1

1.2. Problem Formulation 2

1.3. Scope 2

1.4. Objective 2

CHAPTER 2 LITERATURE STUDY **4**

CHAPTER 3 RESEARCH METHODOLOGY **6**

3.1. Research Methodology 6

3.2. Dataset Collection 7

3.3. Pre-Processing Data 8

3.3.1. Checking Missing Values 8

3.3.2. Data Normalization 8

3.3.3. Time Series Data Transformation 8

3.4. Dataset Split (Train-Test split) 10

3.5. Modeling (Model Selection & Training) 11

3.5.1. LSTM (Long Short-Term Memory) 11

3.5.2. GRU (Gated Recurrent Unit) 11

3.5.3. BILSTM (Bidirectional LSTM) 11

3.6. Model Evalluation 12

3.6.1. Mean Absolute Error (MAE) 13

3.6.2. Root Mean Squared Error (RMSE) 14
CHAPTER 4 IMPLEMENTATION AND RESULTS 15
4.1. Experiment Setup 15
4.1.1. Hardware Specifications 15
4.1.2. Software Specifications 16
4.1.3. Libraries Used 16
4.2. Evaluation Result 16
4.2.2. LSTM Model Analysis 17
4.2.3. GRU Model Analysis 17
4.2.4. BiLSTM Model Analysis 18
4.3. Discussion 18
4.3.1. Comparison of actual data and LSTM predictions 19
4.3.2. Comparison of actual data and GRU predictions 20
4.3.3. Comparison of actual data and BILSTM predictions 21
CHAPTER 5 CONCLUSION 22
REFERENCES ..... a
APPENDIX ..... b

LIST OF FIGURE

Figure 3.1 Data Collection and Preparation Data 6

Figure 3.2 Model And Evaluation 7

Figure 4.1 Comparison of actual data and LSTM predictions 19

Figure 4.2 Comparison of actual data and GRU predictions 20

Figure 4.3 Comparison of actual data and BILSTM predictions 21

LIST OF TABLE

Table 3.1. Data Time Series 9

Table 4.1. Comparing the Evaluation Results of Three Models 16

CHAPTER 1
INTRODUCTION

1.1. Background

Rice is a staple food for the Indonesian people, making it a strategic commodity in the national food security system. One of the main indicators in assessing the efficiency of the rice farming sector is rice production, namely the amount of harvest per unit of land area (usually expressed in tons per hectare). This production reflects how optimally agricultural land is utilized to produce rice, making it an important parameter in measuring the efficiency and effectiveness of the agricultural system. In the context of food security, increasing production is one of the main strategies to overcome the challenges of limited land and increasing population growth [1].

To address the problem of forecasting national rice production, the method used in this study implements machine learning approaches using deep learning-based models with the Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) models. Both models belong to the Recurrent Neural Network (RNN) architecture specifically designed to handle sequential data such as time series. LSTM and GRU are suitable for remembering information from long periods and detecting nonlinear trends in historical data and are therefore very suitable for agricultural yield forecasting problems with high dependence on seasonal patterns and annual variations. Experiments by Sunendar et al. (2025) concluded that LSTM is better at predicting complex and long-term trends and GRU performs the same predictions with a shorter training time [2].

In this case study, national rice production data obtained from the BPS will be utilized to develop a prediction model using deep learning approaches. The study will implement three types of Recurrent Neural Network (RNN) architectures, namely Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Bidirectional LSTM (BiLSTM), to model time series data.

The dataset will go through various preprocessing techniques such as data cleaning, normalization, and conversion to a sequence by the sliding window technique. The training and testing will be done on the same data set for all models for comparability purposes.

In addition, the evaluation of the models' performance will be based on evaluation criteria like MAE and RMSE. The objective of the research is to evaluate the ability of each model to recognize any patterns in the data of rice production and select the best model for predicting rice production in the country.

1.2. Problem Formulation

This study aims to explore several questions related to predicting national rice production using deep learning approaches, especially LSTM and GRU models. The main questions discussed in this research are:

1. How effective is the LSTM model in predicting national rice production?
2. How efficient is the GRU model when making predictions about national rice production?
3. How efficient is the prediction of national rice production by the BiLSTM model relative to other models like LSTM and GRU?
4. Which LSTM, GRU, BILSTM model provides better predictions for national rice production forecasting?

1.3. Scope

- This study examines deep learning time series forecasting models BiLSTM, GRU, and LSTM as part of an RNN architecture designed to process sequential data and learn from historical patterns. The performance of both models is evaluated using MAE and RMSE.

- The main dataset is multivariate, consisting of key components of rice production and additional factors including rainfall, and average temperature

1.4. Objective

This study aims to develop a prediction model for national rice production in Indonesia using the LSTM and GRU algorithms. This is to ensure the performance of both models on time-series rice production data and whether accurate predictions can be achieved. The accuracy of these projections is intended to provide information to the government and other stakeholders in planning the development of national production and food security policies going forward. This has implications for government planning strategies regarding food logistics, imports, rice stocks,

and distribution. Furthermore, this study also seeks to compare the performance of BiLSTM and determine a predictive econometric climate variable framework on national rice production in terms of accuracy and bias.

CHAPTER 2
LITERATURE STUDY

Musdani et al. [1] reported that the LSTM model performs well in predicting agricultural-related data by effectively learning patterns from historical observations. Although their research focused on rice prices, the model's ability to capture long-term dependencies makes it relevant for production forecasting. Sunendar et al. [2] conducted a comparative study involving ARIMA, LSTM, and GRU models. Their results indicated that LSTM achieved higher prediction accuracy, while GRU offered comparable performance with faster training time due to its simpler architecture.

Gunawan et al. [3] applied LSTM to predict sales in a furniture store with satisfactory results. Although the domains are different, the data characteristics and prediction objectives are similar, namely projecting future values based on historical patterns. Pratama and Sanjaya [4] also used LSTM in the context of sales turnover prediction and found that this model was able to follow existing fluctuation patterns, something that is important in modeling rice production which is also highly volatile. Setiawan et al. [5] combined LSTM with the optimization algorithms Root Mean Square Propagation and Adaptive Moment Estimation, showing that parameter tuning can significantly improve model accuracy, which is very relevant in improving the performance of rice production prediction.

Tombeng and Ardian [6] used a deep learning approach to predict supermarket sales, emphasizing the importance of data preprocessing and normalization in supporting model stability, which also applies to agricultural data. Azhari and Haryanto [7] applied hyperparameter-tuned LSTM to predict garlic prices. Their research showed that LSTM can be flexibly adjusted to suit the characteristics of the time series data used, including production data.

In comparison, Prastyo and Wahyuningrum [8] used linear regression to predict sales. Although this approach is simpler, they found its limitations in handling non-linear and seasonal data, weaknesses that can be overcome with LSTM and GRU approaches. Anwar et al. [9] used the Apriori algorithm to analyze purchasing behavior. Although their work did not focus on prediction, it highlights the importance of extracting patterns from historical data. Putri et al. [10] utilized

backpropagation neural networks to model customer purchasing patterns based on transaction data, which strengthens the evidence that artificial neural networks have high flexibility for various types of predictive data.

Unlike those studies, this project focuses on predicting national rice production as an indicator of agricultural sector performance, crucial for food security. This approach is unique because it addresses national agricultural issues from a land-efficiency perspective and proposes solutions based on machine learning. which has been proven effective in various fields. This study also aims to provide information that can be used by the government and policymakers in planning and evaluating national food programs.

CHAPTER 3
RESEARCH METHODOLOGY

3.1. Research Methodology

The research process was performed in several steps by developing a national rice production forecasting model based on LSTM, GRU, and BiLSTM models. Initially, the process started with data collection regarding the production of rice obtained from the BPS and climatic variable data obtained from BMKG and NOAA during the year 2000 until 2024. The data utilized in this research comprised about 300 entries.

After that, the data is put through a pre-processing step in which the quality of the data is checked for any gaps in the data and normalized using the MinMaxScaler method to make sure that all data is in the same scale.

The data was then converted to a time series with the use of the sliding window method, which enabled its use as the input of the LSTM, GRU, and BiLSTM models. Finally, the data set was split into 80% for training and 20% for testing, where a part of the training data was used as validation data.

The next step is model training using the three algorithms. After training is complete, model performance is evaluated using MAE and RMSE metrics to measure the level of prediction error. Furthermore, the prediction results are visually compared with actual data to assess the model's ability to follow data patterns.

The data collection and preparation process used in this study is illustrated in Figure 3.1. The process begins with dataset collection, followed by a data pre-processing stage to clean and prepare the data for analysis. After that, the dataset is divided into training and testing sets. This sequence ensures that the data is properly prepared before being used in the modeling stage.

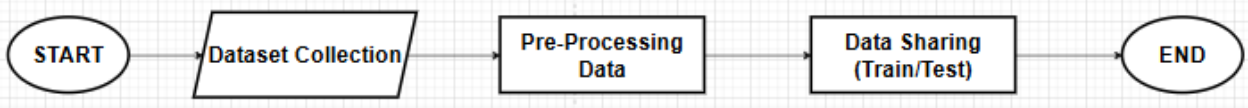


Figure 3.1 Data Collection and Preparation Data

The overall modeling and evaluation process is presented in Figure 3.2. The process begins with model selection, which includes LSTM, GRU, and BiLSTM architectures. Each model is then trained using the prepared dataset.

Furthermore, each architecture has its own mechanism, where LSTM utilizes forget, input, and output gates, GRU employs reset and update gates, and BiLSTM applies bidirectional processing to capture information from both past and future sequences.

After the training process, the models are evaluated using MAE and RMSE metrics. Finally, the performance of all models is compared to determine the most effective model for predicting national rice production.

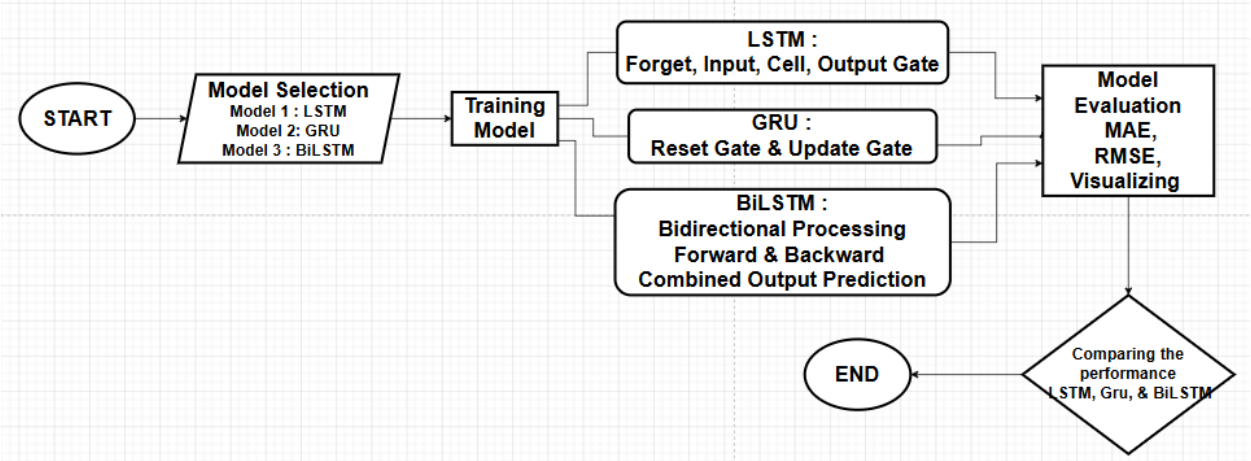


Figure 3.2 Model And Evaluation

3.2. Dataset Collection

This study uses multivariate time series data from 2000 to 2024. In addition to rice production data from the Central Statistics Agency (BPS), climate data such as rainfall (mm), and average temperature (°C) obtained from BMKG and NOAA were also added. The addition of these variables aims to capture relevant external influences on rice production. All data were aligned in time and normalized using MinMaxScaler.

Each data consists of several main attributes, namely:

- Rice production (as target variable)
- Rainfall (mm)
- Average temperature (°C)

3.3. Pre-Processing Data

The pre-processing stage is carried out to ensure data quality before it is used in the modeling process. This stage includes:

3.3.1. *Checking Missing Values*

The data is examined for missing values. If missing values are found, they are handled using linear interpolation based on the previous and subsequent values.

3.3.2. *Data Normalization*

All features are normalized using the method MinMaxScaler to be in the range [0,1]. This aims to speed up the model learning process and increase training stability.

$$X = \frac{X - Xmin}{Xmax - Xmin} \quad [I]$$

where:

- X is the original value,
- Xmin is the minimum value of the feature,
- Xmax is the maximum value of the feature,
- X′ is the normalized value.

3.3.3. *Time Series Data Transformation*

The data is converted into sequence form using the sliding window method, so that it can be used as input for deep learning models. In this study, the sliding window method is applied to multivariate time series data consisting of rice production and climate variables such as rainfall and temperature.

In this research, a **window size of 4** is used, meaning that the model utilizes data from the previous four time steps to predict the value at the next time step.

$$X_t = (x_t^1, x_t^2, x_t^3, x_t^4) \quad \text{[II]}$$

Where :

- x1 = Production
- x2 = Rainfall
- x3 = Temperature

Using a sliding window of size 4, the input-output relationship can be defined as:

$$X_t = (x_t^1, x_t^2, x_t^3)$$

$$y^{(i)} = x_t^1 \quad \text{[III]}$$

This means that the model uses four previous observations to predict the rice production at time *t*.

$$X^{(i)} = \begin{bmatrix} x_{t-4}^1 & x_{t-4}^2 & x_{t-4}^3 \\ x_{t-3}^1 & x_{t-3}^2 & x_{t-3}^3 \\ x_{t-2}^1 & x_{t-2}^2 & x_{t-2}^3 \\ x_{t-1}^1 & x_{t-1}^2 & x_{t-1}^3 \end{bmatrix} \quad \text{[IV]}$$

and the corresponding target:

$$y^{(i)} = x_t^1 \quad \text{[V]}$$

For example, consider the following simplified dataset:

Table 3.1. Data Time Series

Month	Production	Rainfall	Temperature
Jan 2020	50	200	27
Feb 2020	55	180	28
Mar 2020	53	220	27

Apr 2020	60	210	29
May 2020	65	190	30

Table 3.1 shows a sample of the time series dataset used in this study, consisting of rice production, rainfall, and temperature recorded monthly. Rice production acts as the target variable, while rainfall and temperature are used as influencing factors. The data demonstrates fluctuations over time, reflecting the dynamic relationship between climate variables and agricultural production. This structure enables the model to learn temporal patterns and improve prediction accuracy.

Using a sliding window of size 4:

- Input (X):

$$X^{(i)} = \begin{bmatrix} 50 & 200 & 27 \\ 55 & 180 & 28 \\ 53 & 220 & 27 \\ 60 & 210 & 29 \end{bmatrix}$$

- Target (Y):

$$y^{(i)} = 65$$

This sliding window allows the model to detect a pattern from its past observations and then make predictions using this information. By taking into consideration various factors for each window, the model can identify the relationships not only within time but also between time and the impact of climatic variables on rice production.

3.4. Dataset Split (Train-Test split)

The processed dataset is then divided into two parts, namely:

- 80% as training data (training set)
- 20% as test data (testing set)

In addition, some of the training data (20%) is used as a validation set to monitor model performance during the training process.

The division is done chronologically to maintain the characteristics of time series data.

3.5. Modeling (Model Selection & Training)

At this stage, model development and training is carried out using three deep learning algorithms, namely:

3.5.1. *LSTM (Long Short-Term Memory)*

This model was used as the primary model in this study due to its ability to overcome the vanishing gradient problem that often occurs in conventional RNN. LSTM has a gating mechanism consisting of an input gate, a forget gate, and an output gate, which allows the model to selectively control the flow of information. With this mechanism, LSTM is able to retain important information long-term and is very effective in modeling rice production data with seasonal patterns and annual trends.

3.5.2. *GRU (Gated Recurrent Unit)*

GRU is a more straightforward network compared to LSTM. GRU uses a more concise architecture with just two elements: update gate and reset gate. As a result, GRU requires fewer parameters and, thus, trains faster than LSTM. At the same time, despite its simplicity, GRU delivers good results when modeling sequential data.

3.5.3. *BILSTM (Bidirectional LSTM)*

BILSTM is applied as an improvement for LSTM. While LSTM takes input from past to future, BILSTM takes inputs in both directions, past to future and future to past. This technique ensures that the model understands the data better by considering both the past and future of the data, thus capturing more advanced trends of rice production data.

The selection of model parameters in this study is based on dataset characteristics, model complexity, and training stability considerations. Since the dataset consists of approximately 300 data points, a moderate configuration is applied to avoid overfitting while maintaining the ability to capture temporal patterns.

A sliding window size of 4 is used to represent short-term dependencies in the time series data without introducing excessive model complexity. This allows the model to utilize recent historical information effectively.

Furthermore, the number of layers and hidden units are set to moderate values to balance learning capacity and generalization ability. The Adam optimizer is selected due to its adaptive learning rate mechanism and its ability to achieve faster convergence compared to conventional optimization methods. Overall, the parameter configuration follows commonly used practices in deep learning-based time series forecasting and is adjusted according to the size and characteristics of the dataset.

3.6. Model Evaluation

Once the model training process is completed, the next step is to evaluate the model performance in order to measure prediction accuracy. In this study, the evaluation is conducted using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), which are commonly used metrics in time series forecasting. The selection of these metrics is based on the need to assess prediction performance comprehensively, particularly in capturing both average errors and significant deviations.

Mean Absolute Error (MAE) is used to measure the average magnitude of prediction errors in absolute terms, providing a straightforward interpretation of how close the predicted values are to the actual values. This metric is suitable for evaluating overall model performance, as it is not overly sensitive to small variations in the data.

In addition, Root Mean Squared Error (RMSE) is employed to complement MAE, as it assigns a higher penalty to larger prediction errors due to the squaring process. This characteristic is important in the context of rice production forecasting, where large deviations may significantly affect decision-making processes. Therefore, the combination of MAE and RMSE enables a more comprehensive evaluation by capturing both average prediction errors and significant deviations.

3.6.1. *Mean Absolute Error (MAE)*

The MAE formula finds the difference between the actual and estimated data points, and the absolute value is taken. The sum of all error points is then computed and averaged.

MAE is calculated using the formula:

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |Yi - Y^i| \qquad \text{[VI]}$$

where :

- yi = actual value
- y^i = predicted value
- n = number of observation.

MAE is used in this study because it provides a clear interpretation of the average prediction error in absolute terms. This metric reflects how close the predicted values are to the actual values without considering the direction of the error.

In the context of rice production prediction, MAE is suitable for evaluating the general performance of the model, as it represents the average deviation between predicted and actual values. Additionally, MAE is less sensitive to outliers compared to RMSE, making it more stable in representing overall model accuracy.

3.6.2. *Root Mean Squared Error (RMSE)*

Used to provide a larger penalty for significant errors. RMSE is calculated using the formula:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Yi - Y^i)^2} \quad \text{[VII]}$$

where :

- yi = actual value
- y^i = predicted value
- n = number of observations.

RMSE is used to measure the magnitude of prediction errors with a higher emphasis on large deviations. Unlike MAE, RMSE squares the error values before averaging, which results in greater penalization for larger errors.

This characteristic makes RMSE particularly useful in this study, as large prediction errors in rice production forecasting may lead to significant consequences in planning and decision-making.

Therefore, RMSE complements MAE by providing additional insight into how well the model handles extreme errors, ensuring a more comprehensive evaluation of model performance. In addition, a visualization comparison between actual values and predicted results is performed to visually see the model performance.

CHAPTER 4
IMPLEMENTATION AND RESULTS

4.1. Experiment Setup

This research was conducted using a personal computer for development, data processing, and system testing. At this stage, a rice production prediction model was implemented using three Recurrent Neural Network (RNN)-based neural network architectures: LSTM, GRU, and BiLSTM. Experiments were conducted to evaluate the ability of each model to learn time series patterns in national rice production data and the climate variables that influence it.

4.1.1. *Hardware Specifications*

The experiments were conducted with the following hardware specifications:

- Processor: Intel Core i5
- RAM: 8 GB

- Storage: 256 GB SSD

These hardware specifications are considered sufficient to run the computational processes in this study, considering the relatively small dataset size of approximately 300 data points. Furthermore, the training of the deep learning models used in this study, namely LSTM, GRU, and BiLSTM, can still be run successfully on a device with these specifications without requiring high-performance computing devices such as GPUs.

4.1.2. *Software Specifications*

The software environment used in this research is:

- Operating System: Windows 10 64-bit
- Programming Language: Python version 3.11.7
- Development Environment: Visual Studio Code

4.1.3. *Libraries Used*

In this study, the model was implemented using the Python programming language, utilizing several supporting libraries. The main libraries used are:**PyTorch (torch)** to build and train deep learning models such as Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Bidirectional LSTM (BiLSTM). In addition, it is used by NumPy for numerical operations and array manipulation, as well as **Pandas** for processing and analyzing data in tabular form. The data normalization process is carried out using **Scikit-learn (sklearn)**, especially the module MinMaxScaler to change the data scale into a certain range so that the model can learn optimally. To visualize the prediction results and model evaluation, **Matplotlib** displays a comparison graph between actual data and predicted results.

4.2. Evaluation Result

Table 4.1. Comparing the Evaluation Results of Three Models

Model	MAE	RMSE
LSTM	0.358	0.433
GRU	0.396	0.517

BILSTM	0.377	0.462

Table 4.1 presents the comparison of evaluation results for the three models based on Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics are used to measure the prediction accuracy of each model, where lower values indicate better performance.

Based on the results, the LSTM model achieves the lowest MAE value of 0.358 and RMSE value of 0.433, indicating that it provides the most accurate predictions among the three models. This suggests that LSTM is more effective in capturing the underlying patterns of the time series data.

The BiLSTM model shows competitive performance with MAE of 0.377 and RMSE of 0.462, performing better than GRU but slightly below LSTM. Meanwhile, the GRU model records the highest error values, with MAE of 0.396 and RMSE of 0.517, indicating lower prediction accuracy compared to the other models.

Overall, these results demonstrate that LSTM outperforms GRU and BiLSTM in terms of prediction accuracy, while BiLSTM serves as a competitive alternative and GRU provides relatively lower performance.

4.2.2. *LSTM Model Analysis*

An MAE value of 0.358 and an RMSE value of 0.433 was obtained using the LSTM model. From the low values of errors generated by the LSTM model, it can be concluded that the model is accurate in forecasting the production of rice. The lower RMSE indicates that there are no deviations in the predictions of the model from the real data.

4.2.3. *GRU Model Analysis*

According to the results, the GRU gave a Mean Absolute Error of 0.396 and a Root Mean Squared Error of 0.517. In comparison with the LSTM model, the errors have higher figures, meaning that the prediction accuracy of the GRU model is not very high. The higher value of RMSE implies that the model gives relatively bigger errors for some of the data points.

4.2.4. *BiLSTM Model Analysis*

The BiLSTM model had an MAE of 0.377 and RMSE of 0.462. This means that BiLSTM performed better than the GRU but slightly worse than LSTM. With the bi-directional approach, the network has more opportunities for capturing context-based data, leading to improved prediction accuracy. But in this particular case, it was not able to significantly outperform the traditional LSTM model.

According to the result obtained, the LSTM model performs better, with the smallest value for MAE and RMSE. The BiLSTM model comes second, performing well due to its bidirectional processing ability. Finally, the GRU model has the worst performance compared to the other two models.

4.3. Discussion

Visualizations are conducted using the test dataset to evaluate the model's performance in predicting unseen data. The following section presents a visual comparison between actual and predicted values for the LSTM, GRU, and BiLSTM models.

The visualization results indicate that the LSTM model is able to capture the underlying pattern of the data more accurately compared to the other models. This finding is consistent with previous studies, which show that LSTM is effective in modeling long-term dependencies in time series data due to its memory cell mechanism [11].

In contrast, the GRU model demonstrates lower prediction accuracy. Although it offers a simpler architecture and faster training process, its ability to capture complex temporal relationships is generally more limited. This observation is supported by prior research, which suggests that GRU may perform slightly below LSTM in handling complex sequential data [12].

Furthermore, the BiLSTM model demonstrates competitive performance by incorporating information from both past and future sequences. This bidirectional capability enables the model to capture contextual patterns more effectively. This finding is supported by previous research, which shows that bidirectional recurrent neural networks are able to improve sequence modeling performance by utilizing information from both directions [13]. However, the improvement in

prediction accuracy is not significant enough to outperform the LSTM model, particularly when applied to a relatively limited dataset.

4.3.1. *Comparison of actual data and LSTM predictions*

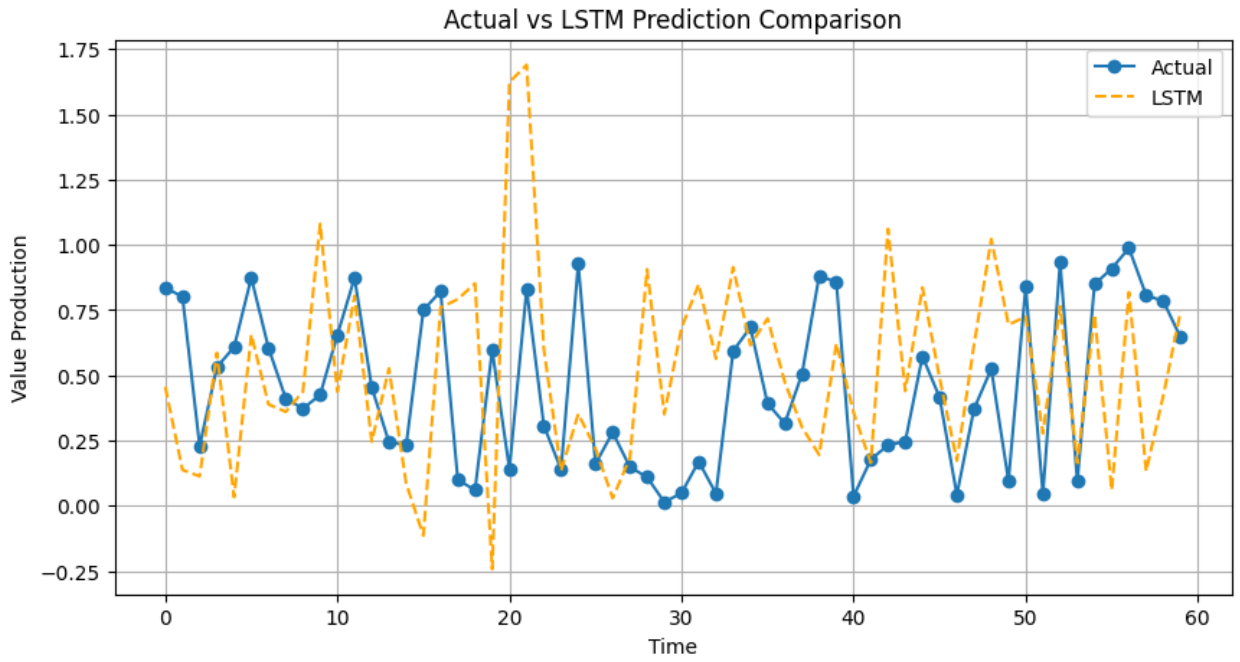


Figure 4.1 Comparison of actual data and LSTM predictions

Based on Figure 4.1, the LSTM model demonstrates a strong ability to follow the pattern of the actual rice production data over time. The predicted values closely align with the actual values across most time steps, indicating that the model successfully captures the temporal dependencies within the dataset.

However, several deviations can still be observed, particularly at certain peak and low points where the predicted values slightly overshoot or undershoot the actual data. These discrepancies suggest that although LSTM is effective in modeling general trends, it may still face challenges in capturing sudden fluctuations or extreme changes in the data.

Overall, the LSTM model provides stable and consistent predictions, which is in line with its lowest MAE and RMSE values among all models. This confirms that LSTM is highly suitable for modeling time series data with complex and long-term dependencies such as rice production.

4.3.2. *Comparison of actual data and GRU predictions*

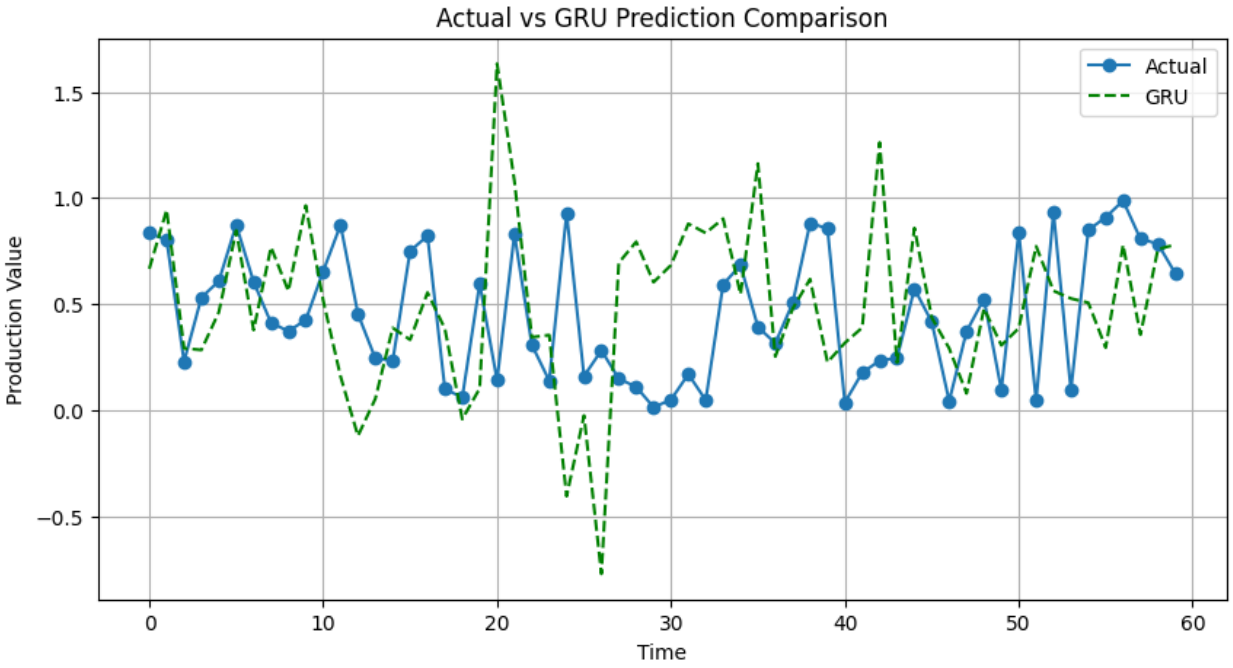


Figure 4.2 Comparison of actual data and GRU predictions

Based on Figure 4.2, the GRU model is able to capture the general trend of the actual data; however, its predictions show greater deviation compared to the LSTM model. The predicted values tend to fluctuate more and are less consistent in following the actual data pattern.

In several instances, the GRU model fails to accurately predict sharp increases or decreases, indicating limitations in capturing complex temporal relationships. This may be due to the simpler architecture of GRU, which uses fewer parameters than LSTM, making it less capable of modeling long-term dependencies in detail.

Despite these limitations, GRU still provides a reasonable approximation of the overall trend, which suggests that it remains effective for simpler time series patterns. However, its higher MAE and RMSE values confirm that its prediction accuracy is lower compared to LSTM.

4.3.3. *Comparison of actual data and BILSTM predictions*

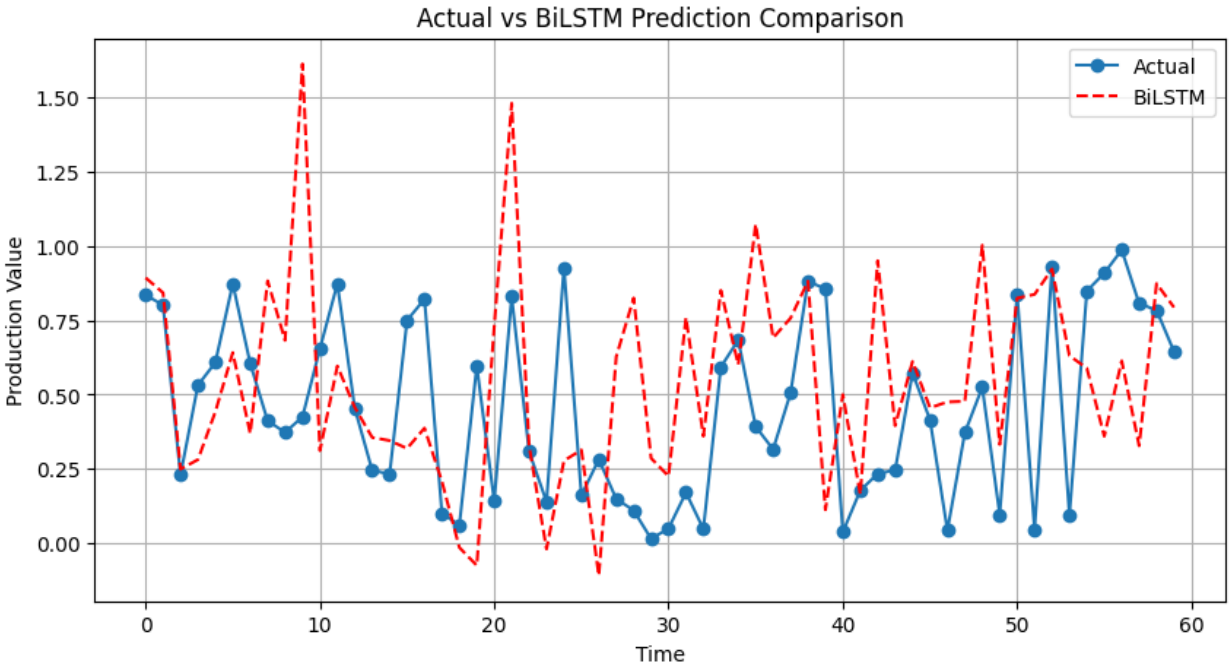


Figure 4.3 Comparison of actual data and BILSTM predictions

Based on Figure 4.3, the BiLSTM model produces predictions that are generally closer to the actual values compared to the GRU model. The predicted line follows the actual data pattern more consistently, particularly in relatively stable regions. This indicates that the bidirectional mechanism helps the model better understand contextual information from both past and future sequences.

However, in some parts of the graph, BiLSTM predictions appear slightly more volatile and occasionally overshoot the actual values. This behavior may indicate a tendency toward overfitting due to the increased complexity of the model.

Although BiLSTM shows improved performance over GRU, the improvement compared to LSTM is not significantly substantial. This is also supported by the evaluation results, where BiLSTM has slightly higher error values than LSTM. Therefore, while BiLSTM is capable of capturing more complex patterns, LSTM remains the most optimal model for this dataset.

Overall, the visualization results support the previous evaluation results, where the LSTM model demonstrated the best performance in following actual data patterns. The BiLSTM model came in

second with fairly good predictive ability, while the GRU model showed a larger deviation. This indicates that model complexity influences its ability to capture time series data patterns.

CHAPTER 5
CONCLUSION

This study aims to evaluate the effectiveness and accuracy of deep learning models, namely Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Bidirectional Long Short-Term Memory (BiLSTM), in predicting national rice production. The analysis provides insight into the capability of each model in handling time series data and identifies the most suitable approach for forecasting national rice production.

Based on the evaluation results, the LSTM model demonstrates the best predictive performance in forecasting national rice production. This is indicated by its lower error values compared to the other models. The results suggest that LSTM is effective in capturing temporal dependencies, particularly long-term patterns in the data.

In contrast, the GRU model exhibits moderate accuracy. Although the model provides benefits such as efficiency and simplicity in design, it does not provide high accuracy when making predictions. It is likely that the GRU model will yield larger errors when compared to the actual data, especially in cases where there are more complex patterns to deal with.

Furthermore, the BiLSTM model demonstrates better performance than GRU and is able to capture data patterns more effectively by utilizing bidirectional processing. This allows the model to incorporate additional contextual information from both past and future sequences. However, despite this advantage, the improvement in prediction accuracy is not significant enough to outperform the LSTM model.

Based on the overall comparison, it can be concluded that the LSTM model provides the best prediction performance for national rice production forecasting. The model consistently

demonstrates higher accuracy and stability compared to GRU and BiLSTM. Therefore, LSTM is considered the most suitable approach for this study, while BiLSTM serves as a competitive alternative and GRU as a more efficient but less accurate model.

In conclusion, the selection of an appropriate deep learning model plays a crucial role in time series forecasting. The findings of this study are expected to support the development of more accurate predictive models, which can contribute to decision-making processes related to national food security.

.

REFERENCES

[1] Musdani VP, Lestari D, Ardaneswari G. Predicting National Rice Prices Using Long Short-Term Memory.*Asian Journal of Management, Entrepreneurship, and Social Sciences*2025; 5: 155–172, https://ajmesc.com/index.php/ajmesc/article/view/1284 (2025).

[2] Sunendar N, Putro HP, Hesananda R. Prediksi Penjualan Aerosol Menggunakan Algoritma ARIMA, LSTM Dan GRU.*INSOLOGY: Journal of Science and Technology 2025*; 4: 113–126, https://www.journal.literasisains.id/index.php/insologi/article/view/4868 (2025).

[3] Gunawan R, Dimiliu MB, Valerine K, et al. ANALYSIS OF FURNITURE STORE SALES PREDICTION USING THE LONG SHORT-TERM MEMORY (LSTM) METHOD.*Journal of Information and Computer Engineering (Tekinkom)*2024; 7: 716–725, https://www.jurnal.murnisadar.ac.id/index.php?journal=Tekinkom&page=article&op=view&path%5B%5D=1511 (2024).

[4] Pratama MB, Sanjaya FI. Prediksi Omset Penjualan Toko Mbak Ning Pasar Cinderamata Menggunakan Long Short Term Memory.*1*2025; 10: 160–167, https://e-jurnal.stmikbinsa.ac.id/index.php/simkom/article/view/580 (2025).

[5] Setiawan Y, Tarno T, Kartikasari P. PREDIKSI HARGA JUAL KAKAO DENGAN METODE LONG SHORT-TERM MEMORY MENGGUNAKAN METODE OPTIMASI ROOT MEAN SQUARE PROPAGATION DAN ADAPTIVE MOMENT ESTIMATION DILENGKAPI GUI RSHINY.*Gaussian Journal*2022; 11: 99–107, https://ejournal3.undip.ac.id/index.php/gaussian/article/view/33994 (2022).

[6] Tombeng MT, Ardian Z. Prediksi Penjualan Supermarket Menggunakan Pendekatan Deep Learning.*CogITo Smart Journal*2021; 7: 160–169, https://cogito.unklab.ac.id/index.php/cogito/article/view/306 (2021).

[7] Azhari IC, Haryanto T. Modeling Of Hyperparameter Tuned RNN-LSTM and Deep Learning For Garlic Price Forecasting In Indonesia.*JOURNAL OF INFORMATICS AND TELECOMMUNICATION ENGINEERING*2024; 7: 502–513, https://ojs.uma.ac.id/index.php/jite/article/view/10714 (2024).

[8] Prastyo HA, Wahyuningrum SE. Memprediksi Dan Menganalisa Penjualan PT. Mitra Asia Synergy Dengan Metode Regresi Linear.*Proxy: Journal of Informatics*2019; 3: 10–19, https://journal.unika.ac.id/index.php/proxies/article/view/3623 (2021).

[9] Anwar MT, Purnomo HD, Novita M, et al. IMPLEMENTASI METODE ASOSIASI APRIORI UNTUK MENGETAHUI POLA BELI KONSUMEN DAN REKOMENDASI PENEMPATAN PRODUK PADA SWALAYAN XYZ.*Dynamic*2020; 25: 29–38, https://www.unisbank.ac.id/ojs/index.php/fti1/article/view/7747 (2020).

[10] Putri N, Sarwani MZ, Hariyanto R. "Implementasi Association Rule Pada Transaksi Penjualan Menggunakan Metode Neural Network Backpropagation.*INTEGER: Journal of Information Technology*9. Epub ahead of print September 30, 2024. DOI: 10.31284/j.integer.2024.v9i2.6398, https://ejurnal.itats.ac.id/integer/article/view/6398 (2024).

[11] Hochreiter S, Schmidhuber J. Long Short-Term Memory. Neural Computation 1997; 9: 1735–1780, https://www.researchgate.net/publication/13853244_Long_Short-Term_Memory (1997).

[12] Cho K, Merrienboer B, Bahdanau D, et al. Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation. Epub ahead of print 3 June 2014. DOI: 10.3115/v1/D14-1179, https://www.researchgate.net/publication/262877889_Learning_Phrase_Representations_using_RNN_Encoder-Decoder_for_Statistical_Machine_Translation (2014).

[13] Climate Data Daily IDN, https://www.kaggle.com/datasets/greegtitan/indonesia-climate.

[14] Indonesia BPS. Produksi Padi¹ dan Beras Menurut Provinsi, 2025 - Tabel Statistik, https://www.bps.go.id/id/statistics-table/3/ZDNaak0yODBUVTlGYW5sa2REUkVUVVY1YVZkbmR6MDkjMyMwMDAw/produksi-padi-sup-1--sup--dan-beras-menurut-provinsi.html?year=2024.

[15] Indonesia BPS. Jumlah Curah Hujan dan Jumlah Hari Hujan di Stasiun Pengamatan BMKG, 2011-2015 - Tabel Statistik, https://www.bps.go.id/id/statistics-table/1/MTk1OSMx/jumlah-curah-hujan-dan-jumlah-hari-hujan-di-stasiun-pengamatan---bmkg--2011-2015.html.

APPENDIX

CODING LOAD DAN GABUNG DATA

```
1. import pandas as pd
2.
3. produksi_path = "Produksi_Padi_Bulanan_2000-2024.csv"
4. produksi_df = pd.read_csv(produksi_path)
5.
6. print("=== Preview: Produksi Padi Bulanan (2000–2024) ===")
```

```
print(produksi_df.head())
print("\nInfo Produksi:")
print(produksi_df.info())

iklim_path = "Dataset_Iklim_Bulanan_2000-2024.csv"
iklim_df = pd.read_csv(iklim_path)

print("\n=== Preview: Dataset Iklim Bulanan (2000-2024) ===")
print(iklim_df.head())
print("\nInfo Iklim:")
print(iklim_df.info())

data_merged = pd.merge(produksi_df, iklim_df, on=["Tahun", "Bulan"], how="inner")

print("=== Data Gabungan Produksi + Iklim ===")
print(data_merged.head())

print("\nInfo Data Gabungan:")
print(data_merged.info())

print("\nJumlah Nilai Kosong per Kolom:")
print(data_merged.isnull().sum())

data_merged = data_merged.fillna(method='ffill')

data_merged.to_csv("Data_Gabungan_Produksi_Iklim_Bulanan.csv", index=False)
print("\n✅ Data gabungan berhasil dibuat dan disimpan sebagai 'Data_Gabungan_Produksi_Iklim_Bulanan.csv'")
```

CODING PREPROCESSING DAN NORMALISASI

```
import numpy as np
from sklearn.preprocessing import MinMaxScaler

features = data_merged.drop(columns=["Tahun", "Bulan"])

scaler = MinMaxScaler()
scaled_features = scaler.fit_transform(features)

print("=== Data Setelah Scaling (5 baris pertama) ===")
print(scaled_features[:5])

print("\nNilai minimum tiap kolom:", scaler.data_min_)
print("Nilai maksimum tiap kolom:", scaler.data_max_)
```

CODING PEMBUATAN SEQUENCE (SLIDING WINDOW)

```
def create_sequences(data, seq_length):
    xs, ys = [], []
    for i in range(len(data) - seq_length):
        x = data[i:i+seq_length]
        y = data[i+seq_length, 0]
        xs.append(x)
        ys.append(y)
```

```
    return np.array(xs), np.array(ys)

seq_length = 4
X, y = create_sequences(scaled_features, seq_length)

print(f"\nUkuran X (input sequences): {X.shape}")
print(f"Ukuran y (target): {y.shape}")

import torch
X = torch.from_numpy(X).float()
y = torch.from_numpy(y).float().view(-1, 1)

print("\nContoh satu input sequence:")
print(X[0])
print("\nContoh target untuk sequence tersebut:")
print(y[0])
```

CODING DEFINISI MODEL (LSTM,GRU,BILSTM)

```
import torch
import torch.nn as nn

input_size = X.shape[2]

class LSTMModel(nn.Module):
    def __init__(self, input_size, hidden_size=64, num_layers=2):
        super(LSTMModel, self).__init__()
        self.lstm = nn.LSTM(input_size, hidden_size, num_layers, batch_first=True)
        self.fc = nn.Linear(hidden_size, 1)

    def forward(self, x):
        out, _ = self.lstm(x)
        out = self.fc(out[:, -1, :])
        return out

class GRUModel(nn.Module):
    def __init__(self, input_size, hidden_size=64, num_layers=2):
        super(GRUModel, self).__init__()
        self.gru = nn.GRU(input_size, hidden_size, num_layers, batch_first=True)
        self.fc = nn.Linear(hidden_size, 1)

    def forward(self, x):
        out, _ = self.gru(x)
        out = self.fc(out[:, -1, :])
        return out

class BiLSTMModel(nn.Module):
    def __init__(self, input_size, hidden_size=64, num_layers=2):
        super(BiLSTMModel, self).__init__()
          self.lstm = nn.LSTM(input_size, hidden_size, num_layers,
                              batch_first=True, bidirectional=True)
          self.fc = nn.Linear(hidden_size * 2, 1)

     def forward(self, x):
```

```
        out, _ = self.lstm(x)
        out = self.fc(out[:, -1, :])
        return out
```

CODING PEMBAGIAN DATA TRAIN DAN DATA

```
train_size = int(len(X) * 0.8)

X_train, X_test = X[:train_size], X[train_size:]
y_train, y_test = y[:train_size], y[train_size:]

print(f"Total data: {len(X)}")
print(f"Data train: {len(X_train)}")
print(f"Data test : {len(X_test)}")
```

CODING TRAINING MODEL

```
def train_model(model, X_train, y_train, X_test, y_test, epochs=300, lr=0.01, name="Model"):
    criterion = nn.MSELoss()
    optimizer = torch.optim.Adam(model.parameters(), lr=lr)

    for epoch in range(epochs):
        model.train()
        output = model(X_train)
        loss = criterion(output, y_train)

        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        if (epoch + 1) % 50 == 0:
            model.eval()
            with torch.no_grad():
                test_pred = model(X_test)
                test_loss = criterion(test_pred, y_test)
            print(f"[{name}] Epoch {epoch+1}/{epochs}, Train Loss: {loss.item():.6f}, Test Loss: {test_loss.item():.6f}")

    return model
```

CODING TRAINING SEMUA MODEL

```
print("\n=== Mulai Training Model (Dengan Data Uji) ===\n")

lstm_model = train_model(
    LSTMModel(input_size),
    X_train,
    y_train,
    X_test,
    y_test,
    name="LSTM"
)

```

```
gru_model = train_model(
    GRUModel(input_size),
    X_train,
    y_train,
    X_test,
    y_test,
    name="GRU"
)

bilstm_model = train_model(
    BiLSTMModel(input_size),
    X_train,
    y_train,
    X_test,
    y_test,
    name="BiLSTM"
)
```

CODING EVALUASI DATA

```
import numpy as np

def evaluate_regression(model, X_test, y_test, name="Model"):
    model.eval()
    with torch.no_grad():
        preds = model(X_test).numpy().flatten()
        actuals = y_test.numpy().flatten()

    mae = np.mean(np.abs(preds - actuals))
    rmse = np.sqrt(np.mean((preds - actuals) ** 2))

    print(f"\n[{name}] Evaluation Result")
    print(f"MAE   : {mae:.3f}")
    print(f"RMSE  : {rmse:.3f}")

return mae, rmse
```